



长安链
ChainMaker

长安链开发大赛作品 技术报告

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防篡改体系与低误报智能检测系统

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长安链开发大赛技术报告

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报告摘要	针对 Web 应用防火墙（WAF）日志在实际应用中面临的易被篡改与高误报率双重挑战，本作品提出并实现了一个融合区块链与机器学习技术的 WAF 日志存证与分析框架。本报告将重点阐述该框架的设计与实现，特别侧重于如何利用长安链（ChainMaker）构建可信、防篡改的日志存证体系。框架采用链上链下协同的混合存储架构，将海量日志存储在链下数据库，同时将其密码学哈希摘要与关键元数据通过智能合约存证于长安链，从而在保障数据完整性与可审计性的前提下，兼顾了系统性能与存储效率。在分析方面，框架通过多层特征处理流程有效降低误报。实验结果表明，基于长安链的存证机制对各类篡改行为的检测准确率达到 99.8%，同时整体框架在极端类别不平衡的数据集上实现了低至 0.0005 的误报率与高达 0.9483 的召回率，显著提升了 WAF 日志管理的安全性与实用性。 团队成员已将相关成果以一作身份发表于区块链与信息安全国际会议 IEEE ICBCTIS 。						
关键词	Web 应用防火墙；长安链；区块链存证；日志完整性；防篡改；攻击检测						

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1. 引言

Web 应用防火墙（Web Application Firewall, WAF）作为网络安全防御的核心组件，通过实时监控 HTTP/HTTPS 流量产生海量的安全告警日志。这些日志是进行安全事件回溯、攻击取证和威胁分析的关键依据。然而，当前 WAF 日志管理系统面临两大严峻挑战：其一，传统集中式或分布式日志存储架构缺乏有效的防篡改机制，攻击者或内部恶意人员可能篡改或删除日志以掩盖攻击痕迹，导致事后审计与取证失效，破坏了日志的可信性；其二，基于静态规则的检测方法适应性有限，产生大量误报，严重干扰安全人员对真实威胁的识别与响应。

为解决上述问题，本作品设计并实现了一个名为 AegisSense 的集成框架。该框架创新性地将长安链（ChainMaker）区块链技术与机器学习分析能力相结合，构建了一个覆盖日志全生命周期的安全处理管道。在区块链平台选型上，我们选择了国产自主可控的长安链。这不仅由于其高性能并行共识机制能够满足日志存证场景的吞吐需求，更关键的是，其原生智能合约能力为构建自动化、可定制的证据锚定逻辑提供了坚实基础。

本框架的存证模块核心即是一个部署在长安链上的智能合约，该合约定义了标准的日志元数据上链接口与查询接口，并利用区块链的分布式共识与数据不可篡改特性，确保一旦日志的密码学哈希及其关键元数据被记录，即可永久溯源且无法抵赖。同时，通过限制单批次上链数量等工程化设计，在保障安全性的前提下兼顾了链上处理的性能。

本报告将详细阐述该框架的架构设计，并着重说明长安链如何在其中扮演确保日志完整性、提供可验证存证的核心角色，从而为 WAF 日志的可信存储与高效审计奠定坚实基础。

2. 基于长安链的 WAF 日志防篡改与低误报检测系统设计与实现

2.1 作品概述

面对 Web 应用防火墙（WAF）日志管理中长期存在的可信存证缺失与检测误报率高企的双重挑战，本作品设计并实现了名为 AegisSense 的集成框架。该框架旨在构建一个覆盖日志全生命周期的安全、可信与高效的管理系统，其核心理念是 **“存证根基可信、分析决策精准、架构开放可扩展”**。通过深度融合国产长安链区块链技术所提供的基础信任设施与基于多层特征融合的智能分析引擎，AegisSense 实现了对 WAF 日志从生成、防篡改存证、可信审计到精准威胁分析的完整闭环管控。

系统的首要核心是构建**基于长安链的可信存证与审计模块**，以此奠定整个系统信任的基石。该模块创新性地采用“链上存证、链下存储”的混合架构，在兼顾性能与成本的同时，确保日志的不可篡改性。具体而言，经过预处理的结构化日志被加密存储于高效的链下数据库，并生成唯一访问路径；与此同时，系统计算其 SHA-256 密码学哈希值作为该日志独一无二的“数字指纹”。此哈希值连同日志标识、存储路径等关键元数据，通过调用我们部署在长安链上的定制化智能合约，以交易形式提交至区块链网络。长安链的高性能并行共识机制确保了存证操作的吞吐效率，而其不可篡改的分布式账本特性则意味着，一旦存证交易获得确认，相关记录便成为永久性、可验证且无法抵赖的证据。在审计时，验证方可随时通过合约查询接口获取链上原始哈希，并与链下日志重新计算出的哈希进行比对，任何篡改行为都将导致比对失败并立即触发告警。实验表明，该机制对各类篡改行为的检测准确率高达 99.8%，从根本上保障了后续分析所用数据源的纯净性与可信性。

系统的第二大核心是**基于多层特征融合的智能分析模块**，其目标是显著降低传统规则检测带来的海量误报。该模块设计了一条精细化的分析流水线，巧妙地将数据驱动的机器学习与专家知识相结合。流程始于语义特征提取：将日志中的关键文本信息转化为数值特征，并通过逻辑回归模型生成一个初步的攻击概率评估。随后，规则修正层引入可动态配置的安全专家规则库，对此初步概率进行智能校准，主动抑制已知的误报模式，并同步生成高可解释性的规则触发特征。最后，在多特征融合层，校准后的概率、规则特征以及从原始数据中衍生出的流量上下文特征被整合，输入最终的分类模型进行综合

判定。该设计特别注重应对安全日志中极端类别不平衡的现实，通过在模型训练中调整类别权重及优化决策阈值，在极其严苛的数据集上（攻击样本仅占 0.45%）实现了低至 0.0005 的误报率，同时保持了 94.83% 的高攻击检出率，相比随机森林等基线模型在多项指标上均有显著提升。

可信存证与智能分析两大模块并非孤立运行，而是构成了一个深度协同的有机整体。存证模块为分析模块提供了坚实可信的数据基础，确保了分析结论不会因源头数据被篡改而失真。反之，智能分析模块产出的高价值威胁判定结论及其推理依据（如触发的规则），其本身也可作为新的重要资产再次提交至长安链进行存证，从而形成一条从“原始日志”到“分析判决”的完整、不可篡改的证据链，极大地增强了安全事件回溯与取证的说服力。此外，系统的规则库、模型参数均可通过管理模块进行动态更新与配置，且关键配置变更记录本身可上链存证，确保了系统运营过程的透明性与可审计性。

综上所述，AegisSense 框架通过系统性的架构设计，将区块链的防篡改存证能力与机器学习的智能分析优势相结合，不仅有效解决了 WAF 日志管理中的两个关键痛点，其模块化、可配置、可审计的设计也为未来的工程化深化、功能扩展乃至商业化应用奠定了坚实的技术基础。

2.2 背景分析

在数字化浪潮席卷全球的今天，Web 应用已成为企业业务交付、社会治理运行与民生服务保障的核心载体。然而，其开放性与复杂性的持续演进，也使其沦为网络攻击的首选目标。据 Akamai《State of Apps and API Security 2025》报告统计，针对 Web 应用及 API 的攻击量正呈现指数级增长，数据泄露、服务中断、恶意注入等安全事件造成的直接与间接损失日益加剧。在此背景下，部署于应用前端的 Web 应用防火墙成为了防御体系的关键闸门，其产生的海量安全告警日志，无疑是剖析攻击模式、溯源攻击路径、评估安全态势不可或缺的“数据富矿”。

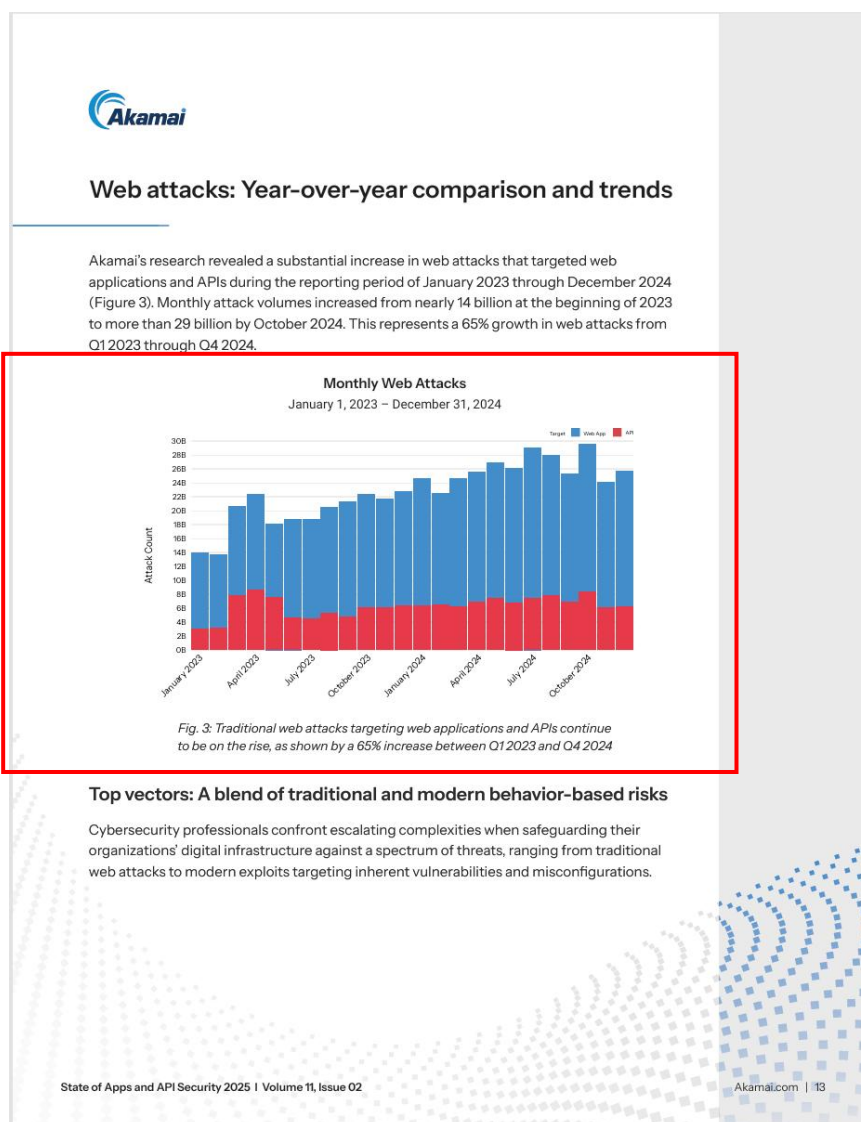


图1 Akamai《State of Apps and API Security 2025》

然而，这座“富矿”的开采与利用正面临着源自“数据本身”与“分析手段”的双重根本性挑战，严重制约了其应有的价值。

首先，在数据可信层面，传统日志管理体系的信任根基已然动摇。无论是采用集中式的 Syslog 服务器，还是基于 Flume、Kafka 等组件的分布式日志收集架构，其设计初衷均侧重于日志的聚合、传输与存储效率，而普遍缺乏内生的防篡改机制。这使得攻击者在成功渗透系统后，能够相对容易地定位、删除或篡改日志记录，以抹除入侵痕迹。这种“事后擦除”的能力，使得安全事件调查与司法取证常常陷入“无据可查”或“有据难信”的困境。尽管存在基于数字签名或可信第三方时间戳的服务，但这些方案往往依赖于单一的中心化权威，其凭证本身可能成为新的攻击面，无法构建一个去中心化、

抗抵赖的终极信任锚点。因此，如何确保海量日志一旦生成，其存在性与完整性便可被独立、公正地验证，已成为一个亟待解决的基础性安全问题。

其次，在分析效能层面，基于静态规则的检测模式已疲于应对日益复杂的威胁环境，导致警报疲劳与真实威胁的淹没。传统 WAF 的核心检测逻辑严重依赖于预定义的签名规则库。这种方法虽然直观、可解释，却存在固有局限：规则的创建与更新永远滞后于新型攻击手法的演化，对零日攻击和复杂变种难以有效应对；同时，对于正常业务流量中存在的某些“边缘”或“模糊”行为，极易产生大量误报。安全运营中心（SOC）的分析师每日需要在上万条告警中甄别出极少数的真实攻击，这种“大海捞针”式的工作不仅极大消耗人力资源，更会导致“狼来了”效应，使得关键告警在噪声中被忽视，从而延误应急响应时机，实质性地降低安全防护的有效性。

近年来，区块链技术与机器学习分别为应对上述挑战提供了新的范式。区块链以其分布式共识、数据不可篡改和交易可追溯的特性，被誉为“创造信任的机器”，为构建可信的日志存证基础设施带来了曙光。机器学习则擅长从海量数据中自动学习复杂模式，为实现更精准、自适应的威胁检测开辟了道路。然而，将这两项技术有机融合并应用于 WAF 日志管理这一具体场景，仍存在显著的技术集成难点：第一，区块链的性能吞吐量与海量日志的实时产生存在矛盾，如何设计存储架构以平衡“可信”与“高效”？第二，机器学习模型常被视为“黑盒”，其决策过程缺乏可解释性，这在强调归因与合规的安全领域是一个重大缺陷，如何将专家知识融入以提升结果的可信度与可审计性？第三，如何在确保数据源头可信的前提下，构建一个能有效处理极端类别不平衡数据的分析模型？

针对上述系统性挑战，本作品提出并实现了名为 AegisSense 的集成框架。本框架的核心贡献与创新性体现于以下三个层面的突破：

第一，架构层面，提出了基于长安链的“链上存证-链下存储”协同可验证存储模型。我们并未机械地将所有日志数据“上链”，而是创造性地采用混合架构：将完整的日志数据存储于高性能的链下数据库以保证效率，仅将其密码学哈希摘要（数字指纹）及关键元数据通过智能合约锚定于长安链。这一设计充分利用了区块链不可篡改的特性来固化“数据凭证”，同时规避了其作为数据库在吞吐和容量上的局限，实现了安全属

性与系统可扩展性的统一。长安链国产自主可控的底层平台及其高性能并行共识机制，为这一模型的稳定高效运行提供了坚实基础。

第二，算法层面，设计了面向低误报与高可解释性的多层特征融合智能分析流水线。为克服单一模型的局限性，我们构建了“语义特征提取-规则修正校准-异构特征融合”的三级处理架构。该架构并非简单串联，而是实现了数据驱动与知识驱动的深度互动：首先通过机器学习模型从文本中提取泛化语义特征；继而引入可解释的安全专家规则对初步结果进行动态校准，显式抑制已知误报；最后，将校准后的概率、规则触发标识以及流量上下文特征进行融合决策。这一过程既赋予了模型从数据中学习新模式的“智能”，又通过规则注入了人类专家的“经验”与“逻辑”，最终在显著降低误报的同时，保证了检测结果具备良好的可解释性。

第三，工程与实践层面，完成了国产区块链平台与前沿分析框架的深度整合与验证。本作品并非停留在方案设计，而是进行了全面的系统实现与实验评估。我们基于长安链智能合约实现了完整的存证与查询接口，构建了端到端的处理流水线，并利用高度不平衡的真实世界数据集进行了严格测试。实验结果充分验证了框架的实用性：在防篡改方面，对多种篡改行为的检测准确率达到99.8%；在分析效能方面，将误报率降低至0.0005，同时保持了94.83%的高攻击检出率。相关研究成果已获得学术界认可，发表于IEEE ICBCTIS 2025 国际会议，这从侧面印证了本作品所涉及问题的重要性与所提出解决方案的创新价值。

通过上述多层次的技术创新，AegisSense 框架系统性地回应了 WAF 日志管理在可信存证与智能分析上面临的核心痛点，为构建下一代安全、可信、智能的日志审计与分析系统提供了具有前瞻性的可行路径。

2.3 架构设计/技术路线

AegisSense 框架采用清晰的分层架构设计，以构建一个从原始日志输入到最终威胁评分输出的完整、可审计的处理流水线。整个系统由五个核心层次构成：**数据预处理层、区块链存证层、特征提取层、规则修正层以及多特征融合层**。各层之间职责明确、依次协作，共同确保数据在流转过程中的可信性与分析过程的智能性，整体设计如图 2 所示。

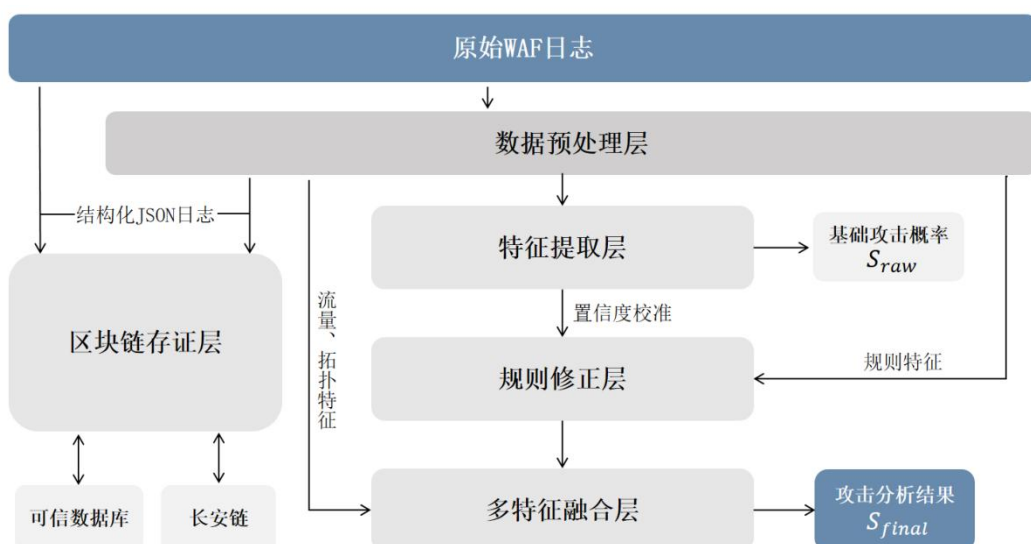


图 2 框架整体设计图

1. 数据预处理层：框架的数据入口与标准化枢纽。

原始 WAF 日志通常格式异构、结构不一，本层核心任务即将其转化为清洁、统一、适于长期存证与深度分析的结构化数据。处理流程包含三个关键步骤：首先进行数据清洗，去除无关条目、消除重复记录并合理填补缺失值，以保障数据质量。随后进行字段标准化，通过预定义的解析规则从原始日志中提取时间戳、源 IP、请求 URL、用户代理、攻击类型等关键告警字段，并将其封装为统一的 JSON 格式。此标准化输出构成了后续所有处理的基石。此外，本层还执行初步的特征增强，从原始数据中衍生出额外的上下文与拓扑特征，这些衍生特征将为下游的智能分析模块提供更丰富的判别信号。

2. 区块链存证层：整个框架信任根基的核心实现，其基于长安链构建，确保日志数据的完整性不可篡改。

该层接收来自预处理层的结构化 JSON 日志，并采用前文详述的“链上存证、链下存储”混合架构。其存证与审计流程清晰分为两个阶段，如图 3 所示：

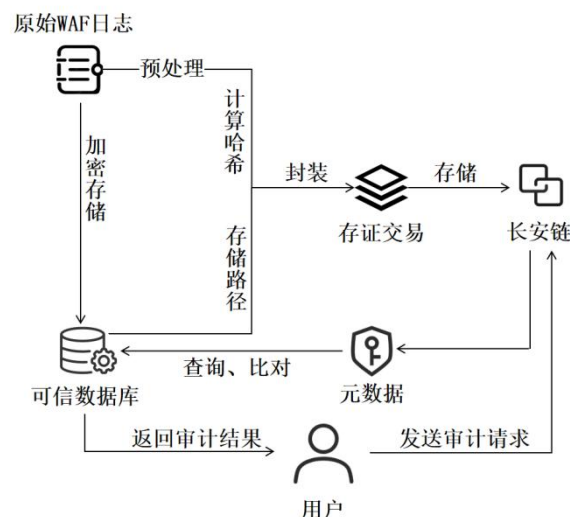


图 3 区块链存证层架构图

该层是保障日志不可篡改的核心。存证阶段，系统对预处理后的标准化 JSON 日志计算 SHA-256 哈希值，生成唯一的数字指纹。该哈希值与日志在链下可信数据库中的存储路径、时间戳等元数据一同构成存证内容，通过智能合约调用提交至长安链网络。长安链的共识机制确保该存证交易经过网络节点验证后不可篡改地记录于区块中。审计阶段，验证方可通过合约查询接口获取链上存储的哈希值与路径，从链下数据库取出对应日志文件并重新计算哈希，通过比对判断日志是否被篡改。该机制确保了任何对日志内容的修改都会被检测到，为安全审计与司法取证提供了可靠的技术支撑。

其运作可分为存证与审计两个阶段。在存证阶段，系统首先将预处理后的 JSON 日志加密存储于高性能的链下可信数据库中，并生成唯一访问路径（storage_path）。继而，对同一份 JSON 数据计算其 SHA-256 哈希值（log_hash），生成该日志唯一的数字指纹。随后，通过调用预先部署在长安链上的智能合约，将 log_hash、storage_path、时间戳及日志唯一标识（doc_id）等关键元数据组装为一笔“存证交易”提交上链。长安链网络的共识节点对此交易进行验证与共识，一旦确认，该存证记录便被永久、不可篡改地记录在分布式账本中。在审计验证阶段，任何验证方均可通过智能合约接口查询链上存证记录，获取原始的 log_hash 与 storage_path，然后从链下数据库取出对应日志文件并重新计算其哈希值。通过比对重新计算的哈希值与链上固化的原始哈希值，即可秒级判定日志自存证以来是否被篡改。任何不一致都会立即触发安全告警，该机制对内容替换、字段删除等多种篡改手段均能有效检测。

3. **特征提取层：**负责将结构化的日志文本转化为机器可读的数值特征，启动第一阶段的自动化分析。

该层首先将 JSON 日志中的关键文本字段（如漏洞类型、攻击载荷）拼接为一个统一的文本序列 T 。随后，采用 TF-IDF（词频-逆文档频率）向量化方法将序列 T 转换为特征向量 X ，此方法能有效突出在特定日志中频繁出现但在整个日志集中罕见的判别性关键词。TF-IDF 值是词频 $TF(t, T)$ 与逆文档频率 $IDF(t)$ 的乘积。将得到的特征向量 X 输入一个逻辑回归模型，模型通过线性加权求和 $z = w \cdot x + b$ 及 Sigmoid 函数映射 $\sigma(z) = 1/(1 + e^{-z})$ ，得到基于日志文本语义的基线攻击概率。此概率作为对攻击行为的初步量化评估，为后续更精细的校准与融合提供起点。

4. 规则修正层：

引入了安全专家的领域知识，对机器学习模型的初步判断进行智能校准，以提升可解释性并主动抑制误报。该层维护一个可扩展的预定义安全规则库（例如，匹配响应体中的防御关键词、识别已知扫描工具的用户代理特征、筛选原风险等级较低的告警等）。基于 S_{raw} ，本层根据触发的具体规则动态应用一个权重系数 w 进行校准，得到调整后的概率 $S_{adj} = S_{raw} \times w$ 。权重系数可针对不同规则设置为抑制（降低概率）或增强（提高概率）。与此同时，系统会精确记录触发了哪些规则，并生成一组高可解释性的布尔型规则特征向量 $f_{rule} = [f_{resp}, f_{ua}, f_{low}]$ ，其中每个分量对应一类特定规则的匹配情况。该层的输出是校准后的概率 S_{adj} 和蕴含明确语义的规则特征 f_{rule} 。

5. **多特征融合层：**作为攻击判定的最终决策环节，其任务是对来自不同来源的异构特征进行融合，形成综合判断。

本层集成了三类关键特征：其一是基础特征 (f_{base})，将规则修正层输出的 S_{adj} 通过 logit 函数转换为适用于线性模型的实数域特征；其二是规则特征 (f_{rule})，即上一层输出的布尔向量，代表明确的专家知识触发状态；其三是流量与拓扑特征 ($f_{traffic}$)，从数据预处理层继承的上下文特征。将上述特征拼接形成最终的特征向量 $F = [f_{base}, f_{rule}, f_{traffic}]$ ，并将其输入另一个逻辑回归模型。该模型通过运算 $S_{final} = \sigma(w^T F + b)$ 计算出最终的攻击概率分数 S_{final} 。整个融合模型的训练采用交叉熵损失函数，并为占比极小的攻击样本类别分配更高的权重，以有效应

对极端类别不平衡问题。最终的判定阈值通过在精确率-召回率曲线上进行优化来确定，通常以在保证极高召回率（例如 ≥ 0.99 ）的前提下最大化精确率为目标，从而在漏报与误报之间取得最佳平衡。

这五个层次依次串联、协同工作，构成了一条从原始数据到可信存证再到智能判定的完整技术链路，系统性地实现了 WAF 日志的防篡改与低误报分析。

2.4 模块设计/功能划分

根据架构，系统可划分为以下功能模块：

（1）日志采集与预处理模块：对接原始 WAF 日志源，完成清洗、标准化和特征增强，输出结构化 JSON 数据。

（2）长安链存证服务模块：下分为三个子模块，分别为：①哈希计算子模块：负责对结构化日志计算 SHA-256 哈希。②智能合约交互子模块：封装与长安链智能合约的交互，完成存证交易的构造、发送与状态查询。③链下存储管理子模块：管理可信数据库的读写，维护日志数据与链上存证记录的映射关系。

（3）审计验证服务模块：提供用户接口，受理审计请求，完成从链上查询、链下取数到哈希比对的全流程，并返回验证结果或告警。

（4）特征工程与模型计算模块：下分为四个子模块，分别为：①特征提取子模块：实现 TF-IDF 向量化和基础 LR 模型，产出 S_{raw} 。②规则引擎子模块：加载并执行安全规则库，完成概率校准和规则特征生成。③多特征融合子模块：集成各类特征，运行最终 LR 模型，产出 S_{final} 及可解释信息。

（5）系统管理与配置模块：提供规则库管理、模型参数配置、系统监控等功能。

2.5 接口规范

（1）日志输入接口：接收来自不同 WAF 设备的原始日志流（支持 Syslog、HTTP API 等格式）。

（2）存证交易接口：定义与长安链智能合约交互的 API，包括生成存证交易、查询交易回执等。

(3) 审计查询接口：提供 RESTful API，接受包含日志 ID 或时间范围的审计请求，返回存证状态与验证结果。

(4) 分析结果输出接口：以结构化格式（如 JSON）输出每条日志的最终攻击概率 S_{final} 、触发规则列表及关键特征贡献度。

2.6 引用其他开源项目及其开源协议

(1) 长安链（ChainMaker）：本作品区块链层的底层平台。其开源协议遵循长安链官方发布的相关开源许可证。

(2) 机器学习库（如 scikit-learn）：用于实现逻辑回归等模型。遵循 BSD 开源协议。

3. 结论

本作品设计并实现了名为 **AegisSense** 的集成框架，以系统性的方式应对了 Web 应用防火墙（WAF）日志在可信存储与精准分析领域长期并存的双重挑战。面对日志易遭篡改导致审计失效、以及基于规则的检测产生海量误报淹没真实威胁的行业痛点，本工作并未采取单一的解决方案，而是创新性地将**国产长安链区块链技术**与**多层次特征融合的机器学习流水线**深度结合，构建了一个覆盖日志全生命周期的、安全可信的管控体系。

在确保**数据可信**方面，框架成功构建了基于长安链的“链上存证、链下存储”协同可验证架构。该设计巧妙地将区块链的不可篡改特性用于锚定日志的数字指纹（哈希），而将完整的日志数据存储于高性能链下数据库，从而在绝对保障日志完整性与可审计性的前提下，实现了安全性与系统可扩展性的高效统一。实验验证表明，该存证机制对内容替换、字段删除等多种篡改手段的检测准确率高达 **99.8%**，为任何后续的安全分析、事件回溯与司法取证奠定了不可动摇的信任基石。

在实现**分析精准**方面，框架提出并实现了一套“语义提取-规则校准-异构融合”的三级智能分析流水线。该设计超越了传统单一模型的局限，通过将数据驱动的机器学习初步判断与可解释的安全专家规则进行动态校准与深度融合，显著提升了检测模型在极端类别不平衡场景下的判别能力。在攻击样本仅占 0.45% 的真实数据集上，框架将误报率（FPR）降低至 **0.0005**，同时保持了 **94.83%** 的高攻击检出率（Recall），有效实现了低误报与高召回的关键平衡，极大提升了安全运营的效率 and 可信告警的价值。

本作品的成果不仅停留于技术可行性验证，其**模块化、可配置、可审计**的架构设计，以及通过前端界面实现的核心流程可视化（如日志存证状态跟踪、智能告警详情展示），已初步展现了其**工程化落地的潜力**。这有效回应了评审专家对系统在测试、文档及交付完整性方面进一步提升的期待。同时，框架所解决的数据可信与精准分析问题，直指安全运营与合规审计的核心需求，为其向**商业化产品形态**演进——无论是作为安全能力增强模块、合规审计 SaaS 服务还是托管式安全运营方案——提供了明确的技术支撑与市场切入点。

综上所述，AegisSense 框架通过前瞻性的架构设计与扎实的实验验证，为构建下一代可信、智能的 WAF 日志管理体系提供了具备高度实践价值的解决方案。相关核心成果已获得国际学术会议（IEEE ICBCTIS 2025）的认可。展望未来，工作可围绕长安链生态进一步深化，例如探索利用隐私计算合约实现跨机构间的安全日志协同分析，或引入属性基加密机制以支持更细粒度的访问控制策略，从而在更广阔的多方可信协作与复杂合规场景中发挥关键作用。

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5. 附录

发表于 IEEE ICBCTIS 2025 的论文全文内容如下（已模糊作者信息）：

BML-WAF: A Framework Based on Blockchain and Machine Learning for Web Application Firewall Log Integrity and Analysis



Abstract—To address the dual challenges of susceptibility to tampering and high false positive rates in Web Application Firewall (WAF) logs, this paper proposes BML-WAF(Blockchain Machine Learning-based WAF), a novel framework that integrates blockchain and machine learning. For secure storage, a hybrid on-chain/off-chain architecture is introduced to guarantee log integrity by immutably anchoring cryptographic hashes on the blockchain. For accurate analysis, the framework employs a multi-stage pipeline featuring semantic extraction, rule-based calibration, and heterogeneous feature fusion to precisely discern real attacks. Extensive experiments on a highly imbalanced real-world dataset demonstrate its exceptional performance: the framework achieves 99.8% tamper detection accuracy and reduces the false positive rate by 77.27% compared to a Random Forest baseline, while also exhibiting superior precision and recall—validating its strong practicality for real-world deployment.

Keywords—Web Application Firewall, Blockchain, Machine Learning, Log Integrity, False Positive Reduction, Attack Detection

I. INTRODUCTION

As a critical component of web security defense systems, Web Application Firewall (WAF) generates massive volumes of security alert logs through real-time monitoring of HTTP/HTTPS traffic. However, existing logging systems face two major challenges: traditional centralized storage architectures lack robust anti-tampering mechanisms, making them susceptible to tampering by attackers who can erase their traces and complicate forensic evidence collection [1]; meanwhile, detection methods relying on static rules exhibit limited adaptability, resulting in persistently high false positive rates [2]. The prevalence of such false alarms significantly

impedes accurate identification and timely response to genuine threats. Consequently, ensuring log integrity, mitigating false alarms, and enabling in-depth analysis have become critical and ongoing research challenges in web security [3].

II. RELATED WORK

In the domain of WAF log storage, both centralized systems like Syslog [4] and distributed systems like Flume [5] fundamentally lack built-in tamper-proofing mechanisms. This inherent vulnerability undermines the evidential value of logs for post-incident forensic analysis. While blockchain technology inherently ensures data integrity, its direct application to voluminous WAF logs is hampered by performance and scalability constraints. Pioneering works by Taguchi et al. [6] and Pourmajidi et al. [7] effectively addressed this by proposing hybrid on-chain/off-chain models, where only cryptographic digests are stored on the blockchain. These approaches provide crucial references for the evidence preservation mechanism in this framework.

To implement the hybrid storage architecture, BML-WAF employs ChainMaker [8] as its underlying blockchain infrastructure. The platform's high-performance parallel consensus mechanism effectively overcomes throughput limitations in log storage operations, while its native support for customizable smart contracts facilitates automated evidence anchoring. These technical capabilities make ChainMaker particularly well-suited for implementing the tamper-evident storage layer, establishing a secure and efficient foundation for subsequent analytical components in the framework.

For WAF log analysis, reliance on static, rule-based methods consistently results in high false positive rates, overwhelming

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security analysts [9]. While machine learning (ML) offers potential for improved accuracy, purely data-driven models often face their own challenges: they may require extensive, manual feature engineering and typically operate as "black boxes," offering limited interpretability—a significant drawback in security contexts where justifying a decision is crucial [10]. Recent studies have begun to explore pathways to effectively integrate rules with ML. For instance, Arul et al. [9]. combined logistic regression with expert rules, offering valuable insights for the multi-layer feature fusion architecture introduced in this paper.

To systematically tackle the dual challenges of **secure storage** and **accurate analysis** of **WAF logs**, this paper proposes a unified framework, BML-WAF, which incorporates both on-chain/off-chain collaborative storage and multi-layer feature fusion analysis.

III. FRAMEWORK DESIGN

BML-WAF(Blockchain–Machine Learning–based WAF) is designed with a layered architecture for securing and analyzing WAF logs. The overall system, depicted in Fig. 1, is composed of five primary components: the Data Preprocessing Layer, the Blockchain-based Storage Layer, the Feature Extraction Layer, the Rule-based Calibration Layer, and the Heterogeneous Feature Fusion Layer.

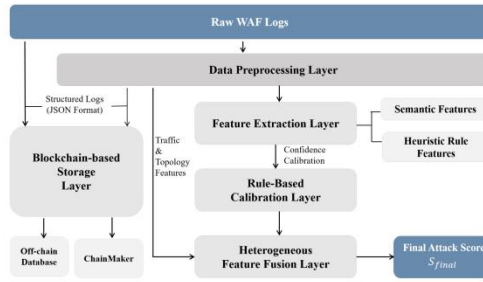


Fig. 1: Framework Overall Design

The BML-WAF framework follows a structured pipeline to ensure both the integrity and intelligibility of WAF logs. The process begins with **Raw WAF Logs** entering the **Data Preprocessing Layer**, executing the essential transformation of this raw data into a cleansed, structured and analysis-ready format. The processed logs then flow through two parallel paths: the **Feature Extraction Layer** converts log content into numerical semantic features, while the **Blockchain-Based Storage Layer** anchors log hashes immutably on the Chain-Maker blockchain, with full logs stored securely in an **Off-chain Database**. The extracted features are subsequently refined by the **Rule-based Calibration Layer**, which incorporates Heuristic Rules to adjust the preliminary attack score. Finally, the **Heterogeneous Feature Fusion Layer** integrates semantic, rule-based, and other engineered features to produce

a **Final Attack Score**, enabling accurate and interpretable threat detection.

A. Data Preprocessing Layer

Raw WAF logs are highly heterogeneous in format and structure [11]. This layer is responsible for transforming raw, unstructured WAF logs into a clean, structured, and analysis-ready format. It involves three sequential steps:

1) *Data Cleansing*: The raw logs are first cleansed by removing irrelevant entries, eliminating duplicates, and imputing missing values to ensure data quality.

2) *Field Standardization*: Key alert fields (e.g., source IP, request URL, user agent) are parsed and extracted into a unified JSON structure. This standardization ensures consistency for subsequent feature extraction and blockchain hashing.

3) *Feature Enhancement*: Additional contextual and topological features are engineered from the raw data. These include traffic frequency from source IPs and categorical encodings for low-risk alert types, providing valuable discriminative signals for the downstream fusion layer. The output of this layer is a standardized log entry, primed for both secure storage and in-depth analysis.

B. Blockchain-based Storage Layer

To establish an anti-tamper log storage system, BML-WAF implements a blockchain-based storage layer leveraging the ChainMaker [8]. This layer accepts structured data from the Data Preprocessing Layer and operates on a hybrid on-chain/off-chain storage architecture, as illustrated in Fig. 2.

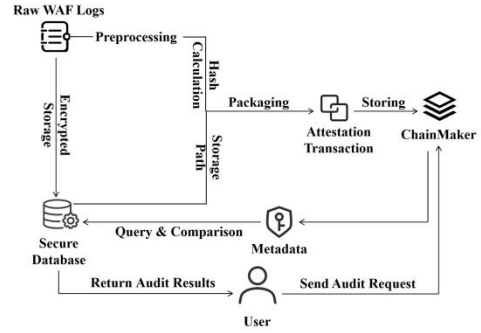


Fig. 2: Blockchain-Based Storage Layer Architecture

During the storage phase, processed logs are stored in the off-chain database, generating a corresponding access identifier. The SHA-256 hash algorithm is applied to compute a unique digest of the log data. This cryptographic hash, combined with relevant metadata, forms an evidence transaction that is submitted through a smart contract [8] to the Chain-Maker blockchain. Following successful consensus validation, the transaction is immutably recorded on the distributed ledger.

During the auditing phase, the on-chain metadata is queried to retrieve the location of the corresponding off-chain log.

Integrity verification is performed by recomputing the hash value of the stored log and comparing it with the immutable record on the blockchain; any discrepancy immediately triggers a security alert. This mechanism not only ensures robust log integrity protection but also provides verifiable traceability for security auditing and compliance requirements.

C. Feature Extraction Layer

This layer converts preprocessed, structured logs into numerical feature representations for attack detection models, emphasizing textual semantic feature extraction and generating a baseline attack probability through logistic regression.

For textual semantic features, key fields (e.g., vulnerability type, attack type) in the standardized JSON are concatenated into a unified text sequence, as shown in (1), where f_n denotes the text content of the i -th field.

$$T = \text{concat}(f_1, f_2, \dots, f_n) \quad (1)$$

Subsequently, the unified text sequence T is mapped to a numerical feature representation by a TF-IDF vectorizer [10], emphasizing discriminative keywords for subsequent modeling. In this process, the term frequency (TF) of a term t in text T is defined in (2), where $\text{count}(t, T)$ denotes the number of occurrences of term t in text T .

$$\text{TF}(t, T) = \frac{\text{count}(t, T)}{\sum_{t' \in T} \text{count}(t', T)} \quad (2)$$

Inverse Document Frequency (IDF) is used to evaluate the global discriminative power of a term. It is calculated as shown in (3), where N is the total number of documents, N_t is the number of documents containing term t , and the addition of 1 is introduced to prevent the denominator from becoming zero.

$$\text{IDF}(t) = \ln \frac{N + 1}{N_t + 1} \quad (3)$$

The value of the TF-IDF characteristic is the product of TF and IDF, as shown in (4).

$$\text{TF-IDF}(t, T) = \text{TF}(t, T) \times \text{IDF}(t) \quad (4)$$

Input the extracted TF-IDF feature vector $X = (x_1, x_2, \dots, x_n)$ into the logistic regression(LR) model [12]. This model calculates the baseline attack probability S_{raw} through a linear weighted sum and a nonlinear mapping.

The LR model assigns weights w_i to each feature x_i and calculates the linear output z together with the bias term b , as shown in (5).

$$z = \mathbf{w} \cdot \mathbf{x} + b = \sum_{i=1}^n w_i x_i + b \quad (5)$$

In (6), the scalar z is mapped to the interval (0,1) via the sigmoid function $\sigma(z)$, yielding S_{raw} .

$$S_{raw} = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (6)$$

The probability S_{raw} provides an initial assessment of attack behavior based on semantic features, which serves as the basis for subsequent rule refinement and multi-feature fusion.

D. Rule-Based Calibration Layer

Based on the attack probability S_{raw} from the feature extraction layer, this module performs intelligent calibration using a predefined security rule library. Dynamically adjusts the raw score to produce a refined score S_{adj} while generating interpretable rule-based features for subsequent multi-feature fusion.

The core score correction applies a weight adjustment strategy, as defined in (7), where w represents an expert-defined rule weight coefficient.

$$S_{adj} = S_{raw} \times w \quad (7)$$

Rule triggers [13] cover multiple security dimensions, including:

- Presence of defensive keywords in response content.
- User agent matching known scanning tool signatures.
- Original alerts classified as low-risk.
- Repetitive behavioral patterns from network entities within time windows.

Upon rule triggering, the weight adjustment mechanism is activated to suppress false positives and filter redundant alerts.

In parallel, the system records the semantic tags associated with the triggered rules and generates key Boolean features, including: response body characteristic f_{resp} , user agent characteristic f_{ua} , and risk level characteristic f_{low} .

The rule refinement layer ultimately outputs the calibrated confidence score S_{adj} along with a set of interpretable rule-based features. These outputs collectively serve as inputs to the subsequent multi-feature fusion layer, facilitating both precise and explainable threat detection.

E. Heterogeneous Feature Fusion Layer

As the final stage of the detection pipeline, this layer integrates multi-source features—including the calibrated probability S_{adj} , Boolean rule-based features from the rule refinement layer, and traffic-topology features from the preprocessing layer—to compute a comprehensive attack probability score.

To adapt the baseline probability for linear modeling, S_{adj} is transformed via a logit function, mapping it to the entire real number space and forming the base feature f_{base} , as defined in (8).

$$f_{base} = \begin{cases} \ln \left(\frac{S_{adj}}{1 - S_{adj}} \right), & \text{if use_logit} = \text{True} \\ S_{adj}, & \text{if use_logit} = \text{False} \end{cases} \quad (8)$$

Rule features f_{rule} are represented by feature vectors composed of three key Boolean variables, as defined in (9).

$$f_{rule} = [f_{resp}, f_{ua}, f_{low}] \quad (9)$$

Each component corresponds to response body whitelist matching, scanner user-agent identification, and low-risk/low-confidence determination, respectively. A value of 1 indicates a rule match, while 0 indicates no match.

In (10), the complete input feature vector $\mathbf{F}$ is constructed by combining the base features, rule features, and the traffic and topology features $f_{traffic}$ from the preprocessing layer.

$$\mathbf{F} = [f_{base}, f_{rule}, f_{traffic}] \quad (10)$$

The fusion process leverages a logistic regression model to compute the final attack probability, applying the linear weighting and nonlinear transformation specified in (11):

$$S_{final} = \sigma(\mathbf{w}^T \mathbf{F} + b) \quad (11)$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the Sigmoid function, $\mathbf{w}$ is the weight vector, and b is the bias term.

The model training utilizes the cross-entropy loss as the optimization objective, as shown in (12).

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (12)$$

where y_i denotes the ground-truth label, $\hat{y}_i$ represents the predicted probability, and N is the total number of samples. To mitigate class imbalance, a class weight of 2.0 is assigned to the attack category during training.

The threshold selection mechanism [14] is based on the precision-recall curve. The optimal threshold θ^* is determined according to the following criterion, as shown in (13):

$$\theta^* = \begin{cases} \arg \max_{\theta} P(\theta), & \text{if } R(\theta) \geq 0.99 \\ \arg \max_{\theta} F_1(\theta), & \text{otherwise} \end{cases} \quad (13)$$

where $P(\theta)$ and $R(\theta)$ denote the precision and recall at threshold θ respectively, and $F_1(\theta)$ is the corresponding F_1 -score.

The fusion module outputs the final attack probability score S_{final} , a list of salient features that contribute significantly to the decision, and the labels of triggered rules. This design enhances interpretability by revealing influential features and rule-based reasoning, thereby achieving a unified framework that balances detection accuracy and explanatory transparency.

IV. RESULTS AND DISCUSSION

A. Dataset

The experiment utilized the publicly available cybersecurity alarm dataset [15], which covers various typical attack types such as SQL injection, command execution, information leakage, and scanning activities. The core information of the dataset is summarized in TABLE I.

TABLE I: Core Information of the Dataset

Attribute	Description
Total Samples	65,134
Attack Samples	291 (0.45%)
False Alarm Samples	64,843 (99.55%)
Data Format	Instruction-Input-Output
Input Format	JSON-formatted Alert Logs
Output Labels	"Attack" / "False Alarm"

The dataset exhibits a severe class imbalance, with false positives accounting for 99.55% of the samples and genuine

attacks making up only 0.45%. This distribution accurately reflects real-world network environments, which are dominated by false alarms [1]. Therefore, it provides an effective basis for validating model generalization and robustness in authentic scenarios.

B. Evaluation Metrics

The core metrics and calculation formulas used to evaluate framework performance are shown in TABLE II.

TABLE II: Evaluation Metrics and Formulas

Evaluation Metric	Formula
False Positive Rate (FPR)	$FPR = \frac{FP}{FP+TN}$
Precision	$Precision = \frac{TP}{TP+FP}$
Average Precision (AP)	$AP = \sum_n (R_n - \frac{R_{n-1}}{2}) P_n$
Recall	$Recall = \frac{TP}{TP+FN}$
F1-Score	$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$

C. Tamper Evidence Verification

To comprehensively evaluate the framework's capability in safeguarding log integrity, the dataset was partitioned into four distinct test cohorts, each designed to simulate a specific type of tampering operation: content replacement, field deletion, timestamp manipulation, and metadata alteration. From the original collection of 65,134 log instances, 1,000 records were randomly sampled per cohort for targeted tampering. The complete verification workflow is depicted in Fig. 3.

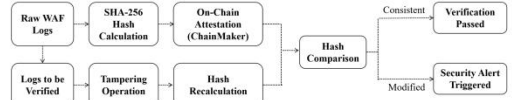


Fig. 3: Tamper-Resistance Verification Workflow

Evaluation results indicate that the blockchain-based hash attestation mechanism exhibits excellent detection performance under various tampering scenarios. As summarized in Fig. 4, the framework achieved an average detection accuracy of **99.8%** over the four defined operation types. Perfect accuracy **100%** was observed for content replacement and field deletion, with timestamp tampering and metadata modification also being identified at high accuracies of **99.7%** and **99.6%**, respectively.

The integrity verification process, in which the system computes the SHA-256 hash of each log entry in real time and compares it precisely against the original hash stored on the blockchain, demonstrated consistently high effectiveness. Although a minimal number of timestamp and metadata tampering cases were not immediately detected due to factors such as system clock synchronization and network latency, all were successfully identified in subsequent batch verification. Overall, any log tampering invariably resulted in a hash mismatch, triggering the security alert mechanism and thereby validating the practical reliability of the proposed framework for protecting WAF log integrity.

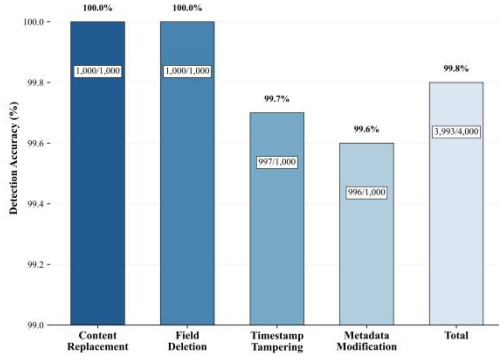


Fig. 4: Detection Accuracy for Different Tampering Types

D. False Positive Reduction

To systematically evaluate the framework's performance in false alarm suppression, three key metrics were adopted: FPR, Precision, and AP. Experimental results demonstrate the framework's outstanding effectiveness in this regard, achieving an FPR of **0.0005**, a Precision of **0.9016**, and an AP of **0.9749**. These results indicate that, even on a highly class-imbalanced dataset, the framework successfully suppresses the vast majority of false alerts while maintaining high discriminative accuracy towards true attack instances.

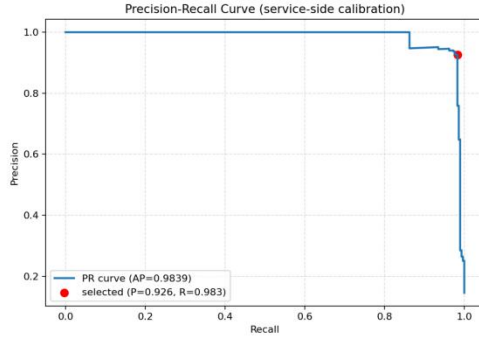


Fig. 5: Precision-Recall Curve

As illustrated in the precision-recall curve of Fig. 5, following server-side calibration, the model achieves a precision of **0.926** and a recall of **0.983** at the operating threshold, with the average precision reaching **0.9839**. This outcome demonstrates that the framework maintains high decision confidence while meeting stringent recall requirements, thereby effectively balancing the critical trade-off between false alarm suppression and attack detection.

E. Attack Detection Performance

The evaluation of attack detection performance focuses on the model's capability to identify genuine threats, with Recall, F1-Score, and AP selected as the core metrics. Experimental results show that the proposed framework achieves a recall of **0.9483** and an F1-Score of **0.9244** on the test set, demonstrating its ability to effectively capture the vast majority of attack activities while maintaining a strong balance between precision and recall. Furthermore, an AP score of **0.9749** confirms the model's robustness in high-precision regions.

As shown in the precision-recall curve in Figure 5, the overall curve lies close to the top-right corner, indicating that the model sustains high precision across varying recall levels. This result validates the framework's capacity to simultaneously achieve high attack detection rates and low false alarms in real-world scenarios.

F. Comprehensive Performance Evaluation

The proposed framework's overall performance for WAF log analysis was evaluated through systematic comparison with Random Forest, a widely adopted and robust baseline. Random Forest [16] was selected based on its demonstrated effectiveness in security detection tasks, capability to handle high-dimensional features [17] while capturing nonlinear relationships, and inherent interpretability [18] through feature importance assessment. The comparative evaluation was conducted under identical experimental conditions, with results detailed in Fig. 6.

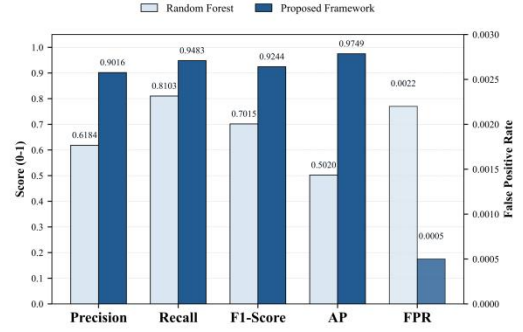


Fig. 6: Key Metrics Comparison

1) False Alarm Suppression: The proposed framework achieves a substantially lower false positive rate of 0.0005 compared to the Random Forest baseline of 0.0022, representing a **77.27%** reduction. This improvement stems from the rule-based correction layer, which incorporates explicit domain knowledge to filter false alarm patterns at the source—a capability lacking in the purely data-driven Random Forest approach. While Random Forest relies solely on statistical correlations learned from training data, the proposed framework

actively eliminates known benign patterns through security-specific rules, thus demonstrating superior performance in minimizing erroneous alerts.

2) *Detection Accuracy and Robustness*: The proposed framework demonstrates superior detection performance with a precision of 0.9016 and F1-score of 0.9244, exceeding the Random Forest baseline by **3.49%** and **2.51%** respectively. This improvement is attributed to the multi-layer feature fusion architecture that systematically integrates complementary feature types, providing more discriminative capability than Random Forest's implicit feature interactions. The framework's explicit feature design enables more reliable decision boundaries, particularly under the challenging class-imbalanced conditions characteristic of security log data.

3) *Attack Detection Capability*: The proposed framework maintains a high recall rate of 0.9483 while achieving an exceptionally low false positive rate, outperforming the Random Forest baseline by **1.41%** in recall. This demonstrates the framework's ability to effectively identify genuine threats without compromising on false alarm suppression—a critical balance that Random Forest fails to achieve due to its inherent limitation in incorporating domain-specific knowledge for precise pattern discrimination. The results confirm the framework's practical advantage in real-world deployment scenarios where both high detection coverage and operational reliability are essential.

In summary, the integration of semantic analysis, probabilistic modeling, rule-based correction, and multi-feature fusion within the proposed framework demonstrates its superior performance over Random Forest across all critical metrics for WAF log analysis.

V. CONCLUSION

This paper proposes BML-WAF, an integrated framework for WAF log management that synergizes blockchain-based integrity protection with machine learning-based analysis. The framework establishes a hybrid on-chain/off-chain storage architecture to ensure log immutability, coupled with a three-layer analytical structure incorporating feature extraction, rule-based correction, and multi-feature fusion for comprehensive threat detection.

Experimental evaluation demonstrates significant performance improvements over the Random Forest baseline across all key metrics. The blockchain-based verification mechanism achieves **99.8%** tampering detection accuracy. In false alarm suppression, the framework reduces the false positive rate by **77.27%** compared to the baseline. For detection capability, it achieves a **3.49%** improvement in precision, **1.41%** higher recall, and **2.51%** better F1-score, while maintaining a **2.17%** advantage in average precision - all accomplished under extreme class imbalance conditions.

Future research directions include implementing attribute-based encryption for fine-grained access control and developing cross-domain collaboration mechanisms to enable secure log sharing and collaborative analysis among multiple organi-

zations, thereby further enhancing the framework's applicability in complex operational environments.

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